

SYLLABUS :-

Review - Discrete time systems, linear time invariant (LTI) systems and important properties. Fourier Transform and Laplace transform. Z-transform. Signal flow graphs and digital system representation; Discrete Fourier transform (DFT) and its properties. Fast Unitary transforms. Introduction to transformation matrices in a general form. Computer exercises; Digital filters- FIR and IIR. FIR filters -structure, designs. IIR filters -analog filter design, discretization of analog filter Computer exercises; Multi-rate signal processing - Interpolation and decimation, FIR filter banks; Introduction to least square optimization, signal modeling (AR, MA, ARMA). Normal equation and solution strategy. Applications. Computer exercises; Digital signal processing hardware -discussion of either TMS320CXX based or ADSPXXX based system; Analysis of practical system implementation issues -effect of finite word length. Coefficient quantization in FIR filters. Limit cycles in IIR filters. Round off effects in FFT computation; Digital processing of continuous time signals- sampling, anti-aliasing filter, sample and hold process, reconstruction filter, Computer exercises;